

Optimal decision-making in the water, land and food nexus using artificial intelligence and extreme machine learning

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ABSTRACT

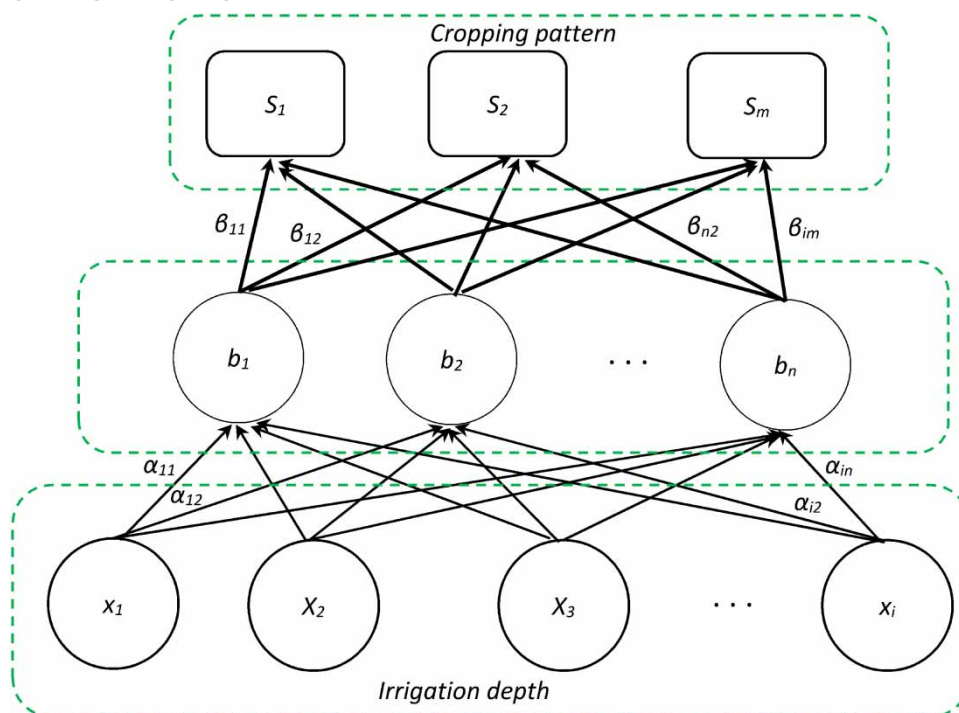
The development of decision-making systems based on artificial intelligence can lead to achieving optimal solutions water-land-food nexus. In this paper, an extreme learning machine model was developed with the objective function of wheat production maximization. The constraints defined for this problem are divided into three categories: technical parameters of production in agriculture, climatic stress on water resources and land limits. The water, land and food nexus was simulated using 23 experimental farms in Henan province during the 2021–2022 cultivation year. Root-mean-square error was used as an error criterion, and Pearson's coefficient was incorporated into the decision-making system as a correlation index of variables. Harvest index, length of the growth period, cultivation costs and irrigation water were the criteria to evaluate the impact of the sustainable model. The harvest index and the length of the growth period showed the highest and lowest correlation with the production rate, respectively. Furthermore, the optimal management of irrigation water and cost had the most significant impact on increasing crop production. The method proposed in this paper can be a virtual cropping model by changing the area under cultivation of a crop in the different farms of a study area, which increases yield production.

Key words: irrigation, production, smart agriculture, sustainable management, wheat

HIGHLIGHTS

- Extreme machine learning has been used to increase wheat production.
- The harvest index, length of growth period, irrigation and cost were the four investigated factors related to the correlation between water and food.
- Modeling based on information and intelligent learning could increase agricultural productivity.

GRAPHICAL ABSTRACT



INTRODUCTION

Food security is the main challenge across the world with a dependency on water, land and managerial factors in agriculture, such as harvest index, growing season and cultivation cost (Mirajkar & Patel 2016). Managerial changes in agricultural activities play a crucial role in irrigation water consumption and crop yield. Considering the wind, precipitation, temperature, soil texture, water quality, drainage, solar radiation, energy, cultivated area, labor force, fertilizing, economy and the length of the growing season were the topics that were widely investigated to predict the yield production in recent years (Rezaei *et al.* 2017; Elavarasan & Vincent 2021). Accurate early crop yield prediction helps market pricing, planning labor, transport and harvest organization (Klompenburg *et al.* 2020).

Frausto-Solis *et al.* (2009) proposed a data-based method for the cropping pattern optimization problem to maximize the net income of every cropping season. The model could provide a framework to evolve efficient cropping patterns and to maximize the benefits for sowing area distribution. Singh (2023) evaluated the criteria related to predicting crop yield and the effects of climate change using machine learning technology in agriculture. Bhimavarapu *et al.* (2023) proposed an improved optimizer function for estimating crop yield accurately with the incorporation of deep learning models along with crop statistics. Manual data were collected between 1901 and 2000 from local agricultural departments for training. Root mean squared error (RMSE) and mean absolute error (MAE) were the main performance metrics to compare the loss after implementing the proposed method and the observed results. The results showed that the proposed model had the advantage of crop yield prediction with accurate prediction. Mokhtar *et al.* (2023) investigated the potential of machine learning models to predict irrigation water requirements in agriculture by evolving multi-scenarios of inputs parameters to indicate the impact of meteorological, crop and soil parameters on crop water demands. Support vector regressor, random forest, deep neural networks (DNN), convolutional neural networks (CNN), long short-term memory (LSTM) and hybrid CNN-LSTM were applied for this subject. This study developed a decision-making to identify the shortages in irrigation water planning.

To protect water resources in arid and semi-arid regions against increasing water demand and climate change, Sepahvand *et al.* (2019) developed a simulation-optimization model for the combined management of surface and groundwater consumption. Minimizing the shortage in supplying irrigation water needs and maximizing the total net profit of agriculture were done using an intelligent planning based on multi-objective genetic algorithm. The results showed the competence of the proposed optimal model to increase the net profit by reducing the negative social and environmental effects. Elavarasan & Vincent

(2021) proposed a regression-based hybrid algorithm for product estimation that showed significant performance over traditional machine learning techniques such as random forest, decision tree, gradient boosting, artificial neural network and deep learning. The new strategy implements reinforcement learning at each selection of a split feature during the tree construction process to efficiently use available examples. They analyze the variable importance measure to select the most fundamental variable for the node-splitting process in model development and promote the efficient use of training data. This integrated hybrid method provides a significant improvement over common strategies, especially for sparse model structures. Virnodkar *et al.* (2020) developed a machine learning framework linked to remotely sensed data to evolve a real-time farm-specific management system to reinforce farmers' ability to make appropriate decisions. Results showed that the use of machine learning techniques combined with remote sensing data could reshape precision agriculture in crop identification, yield prediction and crop water stress assessment.

Advanced decision-making models provide a different structure than simulation systems for prediction and computer advances have created a new space in fast and accurate calculations for extensive multi-dimensional investigation in various sciences, especially water resources management in agriculture (Abolpour *et al.* 2017; Chen *et al.* 2020). Machine learning has been successfully implemented in various fields such as medicine, electronics, etc., and in agriculture, it has been mainly used for irrigation planning, crop yield prediction and estimation, and improvement of soil moisture content (Lamba & Dhaka 2014; Lalehzari *et al.* 2020; Bali & Singla 2022). The application of artificial intelligence (AI)-based models has been successfully used to predict crop yield by considering various factors such as water content, solar radiation, CO₂ fixation and transpiration (Pantazi *et al.* 2016; Chlingaryan *et al.* 2018). It has been reported that machine learning can predict product performance with high accuracy (Govindan & Al-Ansari 2019). In addition, combining machine learning with simulation models can create decision-making systems to determine sustainable management strategies. This study proposes a high-performance computing method and machine learning knowledge to create a novel opportunity to understand, unravel and quantify data-intensive procedures in agricultural operations.

MATERIAL AND METHODS

Artificial neural network

Artificial neural network is a simplified model of the human brain that has the ability to occur non-linear arbitrary processes and combinations for the connection between the input and output of any system. Each neural network consists of three layers: input, hidden and output layers (Elavarasan *et al.* 2018; Elavarasan & Vincent 2020). In each layer, there are a number of processors called neurons, whose responses to the input signal are transferred by transfer functions. Common transfer functions used in artificial neural networks are sigmoid function and hyperbolic tangent function.

Multi-layer perceptron is an artificial neural network that is built on the basis of a computing unit called perceptron. A perceptron takes a vector of inputs with real values and calculates a linear combination of these inputs. If the resulting value is greater than a threshold value, the output of the perceptron will be equal to one; otherwise, it will be equal to negative one. During the training of multilayer perceptron network with the help of error backpropagation learning algorithm, first calculations from the input of the network is done towards the output of the network and then the calculated error values are propagated to the previous layers. At first, the output calculation is done layer by layer and the output of each layer will be the input of the next layer. The neural network models used in this study are multi-layer feedforward network with an error back propagation algorithm and Marquardt–Levenberg learning technique, whose structure is shown in Figure 1. This algorithm is a type of Newton's method, which is designed without the need to calculate the Hessian matrix for fast training with quadratic order.

Extreme learning machine

Learning machine (LM) as a branch of AI provides an effective approach to construct a model for regression and classification of a multivariate and non-linear system (Li & Liu 2022; Karmaoui *et al.* 2023). Without being explicitly programmed, machine learning models learn from training data, i.e., past experience (Virnodkar *et al.* 2020). Extreme learning machine (ELM) is an advanced method for single-layer feed-forward neural network (Figure 2). In this model, the hidden node parameters (input weights, hidden propensities or centers of functions and factors affecting the hidden node) are randomly selected and the output weights are determined analytically using general inverse Moore–Penrose (Pimenov *et al.* 2023). ELM avoids many problems such as stopping criterion, learning rate and learning periods, which are based on learning methods and created by the slope.

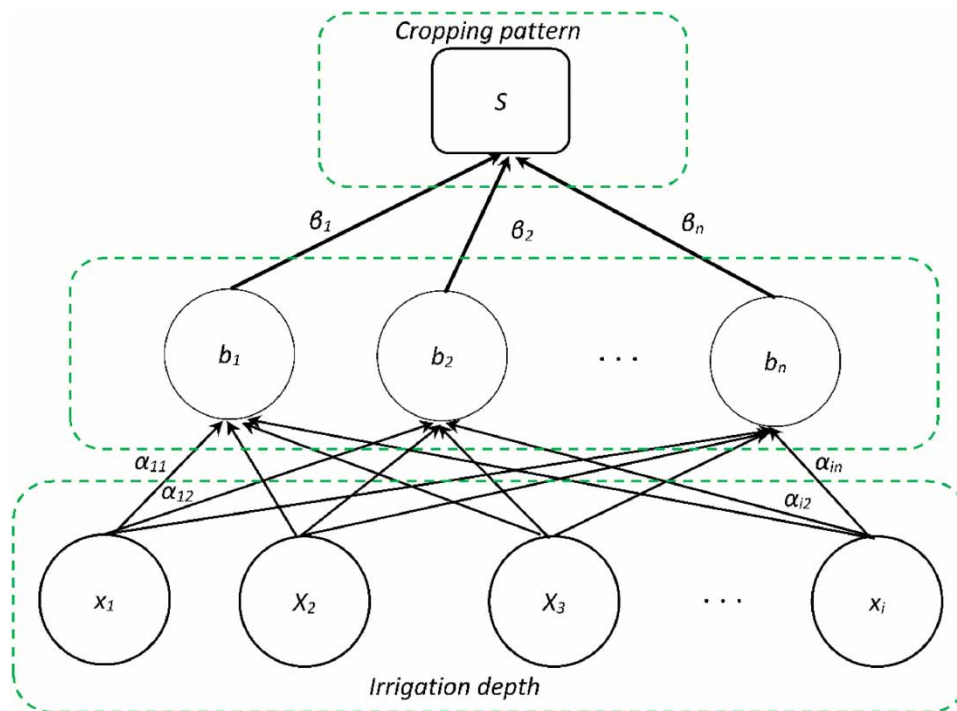


Figure 1 | Learning machine structure as an AI.

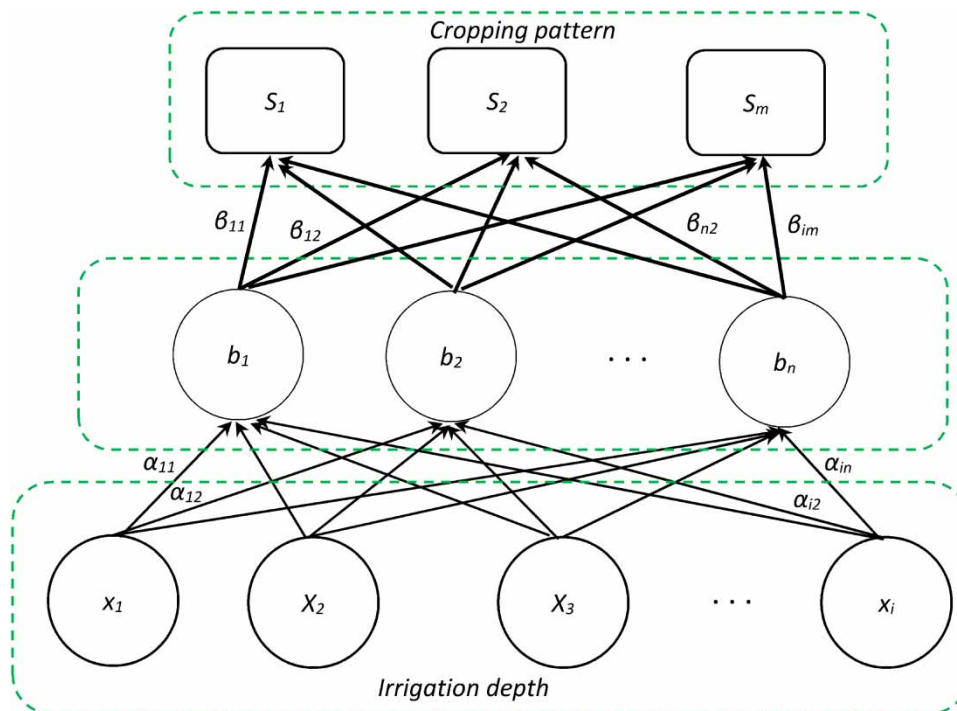


Figure 2 | Conceptual structure of ELM.

Assume that there are N random samples in the training phase, with L number of hidden nodes. Therefore, the single-layer feed-forward neural network model with transfer function can be expressed as follows:

$$\sum_{i=1}^L \beta_i f(x_i) = \sum_{i=1}^L \beta_i f(\alpha_i b_i x_i) \quad \alpha_i = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{im})^T \quad \beta_i = (\beta_{i1}, \beta_{i2}, \dots, \beta_{im})^T \quad (1)$$

where α_i is the input weight vector connected to the hidden layer node, b_i is the bias value of the hidden layer nodes, and β_i is the output weight vector connected to the hidden layer node. The above equation can be written as follows:

$$\sum_{i=1}^L \beta_i f(x_i) = H\beta \quad (2)$$

$$H = \begin{bmatrix} f(\alpha_1 b_1 x_1) & \cdots & f(\alpha_L b_L x_1) \\ \vdots & \cdots & \vdots \\ f(\alpha_1 b_1 x_N) & \cdots & f(\alpha_L b_L x_N) \end{bmatrix}_{N \times L} \quad (3)$$

$$\beta = \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_L^T \end{bmatrix}_{L \times m} \quad (4)$$

$$T = \begin{bmatrix} S_1^T \\ \vdots \\ S_L^T \end{bmatrix}_{N \times m} \quad (5)$$

where H is the output matrix of the hidden layer, β is the weight output matrix, and T is the label matrix.

Based on the conceptual framework, the first step is to determine the random values of the input weights (α) and hidden layer node bias (b) in the training phase. After determining these weights and biases, the output matrix of the hidden layer can be obtained from the input samples. Therefore, the training of a single-layer feed-forward neural network model becomes a least squares solution. After determining the input weights and biases of the hidden layer, the output matrix of the hidden layer can be obtained from the training samples. By introducing the organizing theory for the ELM model, the objective function is expressed as follows:

$$\min L_{ELM} = \frac{1}{2} \|\beta\|^2 + \frac{c}{2} \|T - H\beta\|^2 \quad (6)$$

By solving the above equation in two cases where the examples of the training stage are more or less than the hidden layers, the following equations are obtained, respectively:

$$\beta = H^T / \left(\frac{1}{c} + H^T H \right) \quad (7)$$

$$\beta = TH^T / \left(\frac{1}{c} + H^T H \right) \quad (8)$$

Chithambarathanu & Jeyakumar (2023) reported that these methods enable automatic monitoring of vast domains in agriculture, therefore lowering human error and effort.

Correlation coefficient

Pearson's correlation coefficient (Pcc), which is also called moment correlation coefficient or zero-order correlation coefficient, is used to determine the degree of relationship, type and direction of the relationship between two distance or relative variables or one relative variable and one distance variable (x, y). This correlation coefficient is based on the covariance of two variables and their standard deviation, whose estimates can be used to calculate the Pearson correlation

coefficient. The relationship of this type of correlation coefficient is as follows:

$$P_{cc} = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}} \quad (9)$$

Experimental data

The experimental data were collected from 23 wheat farms located in the dry part of Henan province in China in 2021–2022. The criterion for choosing the fields under cultivation was between 10 and 100 ha. Irrigation information, yield, harvest index, growth period length and farming costs were recorded in one crop season and used as input parameters for model training. The economy in drylands is heavily reliant on agricultural output growth and related agro-industry products (Neto *et al.* 2017; Patrício and Rieder 2018) and the Henan province is predominantly agrarian. Table 1 shows a summary of the collected information.

RESULTS AND DISCUSSION

ELM evaluation

In the first step, the ability of the ELM model to obtain the solution has been evaluated in comparison with AI and LM. In the process of advancing ELM convergence speed calculations, the best solution has been calculated with RMSE = 45 kg/ha

Table 1 | Data source for 23 farms in Henan province

Farm No	Irrigation water mm	Production kg h ⁻¹	Cultivated area ha	Growing season day	Harvest index %	Cultivation cost Yuan ha ⁻¹ season ⁻¹
1	612	4,535	57	164	46.1	10,639.6
2	535	4,325	38	167	42.7	8,337.5
3	478	3,984	24	163	43.2	6,657.5
4	534	4,712	17	172	49.8	9,768.5
5	624	4,311	86	168	47.7	9,457.4
6	519	4,561	26	159	44.6	7,653.1
7	498	4,428	19	163	45.5	7,093.1
8	608	4,610	46	168	45.9	7,715.3
9	586	4,038	28	174	41.3	6,906.4
10	521	4,236	63	178	42.5	8,524.1
11	547	3,729	12	161	39.4	7,777.5
12	514	4,139	90	166	46.6	6,906.4
13	486	4,056	23	158	50.8	6,719.8
14	618	3,826	14	173	37.1	6,719.8
15	602	4,825	94	170	48.7	7,964.2
16	593	4,627	73	165	50.7	8,773.0
17	546	4,218	21	164	42.8	6,968.6
18	506	3,976	32	168	41.3	6,222.0
19	623	4,725	58	157	57.2	8,026.4
20	614	4,804	60	171	52	9,084.1
21	478	4,376	44	166	46.9	8,897.5
22	538	4,008	37	162	48.8	8,586.4
23	632	4,829	26	167	53.7	7,404.2

(Figure 3). The difference of ELM method with LM was 15 kg/ha. Choosing the number of computations to progress the objective function is very effective in calculating the accurate response (Sepahvand *et al.* 2019).

The production simulated by the developed simulation model based on extreme machine learning in the final step is shown in Figure 4. The highest and lowest difference between the observed and simulated values are 97 and 0.4 kg/ha in farms 23 and 20, respectively. The results of the simulation of wheat production showed that the use of field information can lead to acceptable estimates. In simulation models that are developed based on the calculation of plant biomass during the growing season or the estimation of plant production based on water stress, there are logical reasons to show a decrease or increase in yield (Li *et al.* 2018). However, the accuracy of the results refers to the training phase in data-based methods.

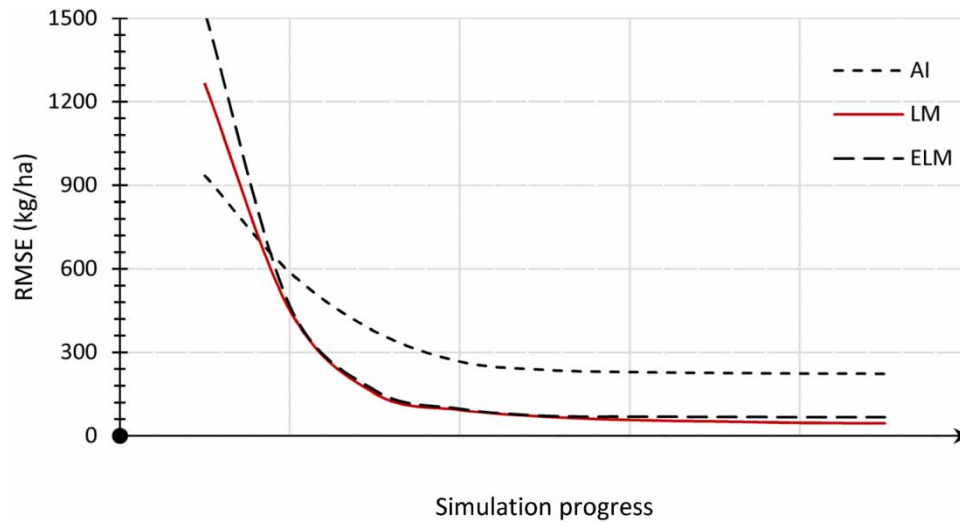


Figure 3 | Root-mean square error changes in simulation progress with 2,000 iterations.

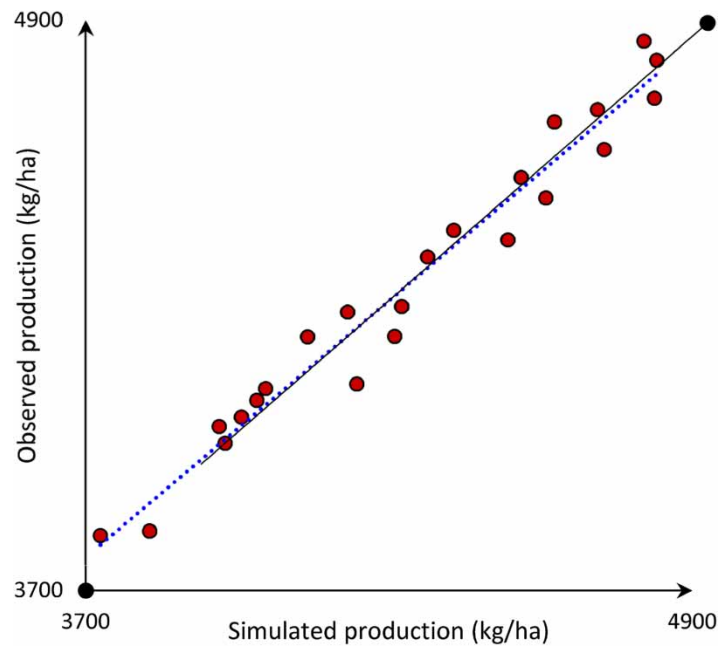


Figure 4 | Observed and simulated production using ELM.

Correlation of cultivated area

Since the purpose of improving the cultivation pattern in this research is to change the area under wheat cultivation for different farms, it is necessary to estimate the correlation between the cultivated area and the yield production. Figure 5 shows the correlation coefficient between cultivated area and production for 23 studied farms. According to the classification illustrated in the figure, 14 farms had a correlation higher than 0.5. It can be concluded that the increase in the cultivated area has led to the improvement of agricultural operations, increased water productivity and the achievement of higher actual yields. Several studies have shown that changing the cultivated area in optimizing the cultivation pattern has increased crop production and food security (Milan *et al.* 2018; Lalehzari & Kerachian 2021).

Sustainable crop production

The sustainable cultivated area calculated by ELM with the observed cultivated area in 2021–2022 is shown in Figure 6. The total amount of land under wheat cultivation in the studied area was 988 ha which was considered a maximum constraint for land allocation. In addition, there has been a restriction that farmers are allowed to change the area under wheat cultivation by a maximum of 40%, and the maximum and minimum change limits were 100 and 10 ha, respectively. As shown in the figure, farm 5, which used 624 mm of irrigation water to produce 4,311 kg/ha, faces a decrease of 14 ha in sustainable conditions. Farm 15, which produces 4,825 kg/ha of crop with 602 mm of water, has reached the upper limit of 100 ha. The optimal irrigation program and the use of effective rainfall potential have improved the efficiency of water consumption on this farm. The maximum increase in cultivated area was obtained for farm 4, which according to Table 1, the total irrigation water consumption in the growing season was equal to 534 mm and resulted in the production of 4,712 kg/ha of wheat. The water productivity in this farm is estimated at 0.88 kg/m^3 , and under optimal conditions, it is recommended that the area under wheat cultivation in the cultivation pattern of this farm be increased from 17 to 28 ha. This increase in cultivated area is equivalent to 40% of the sustainable cultivated area, which was considered a limit.

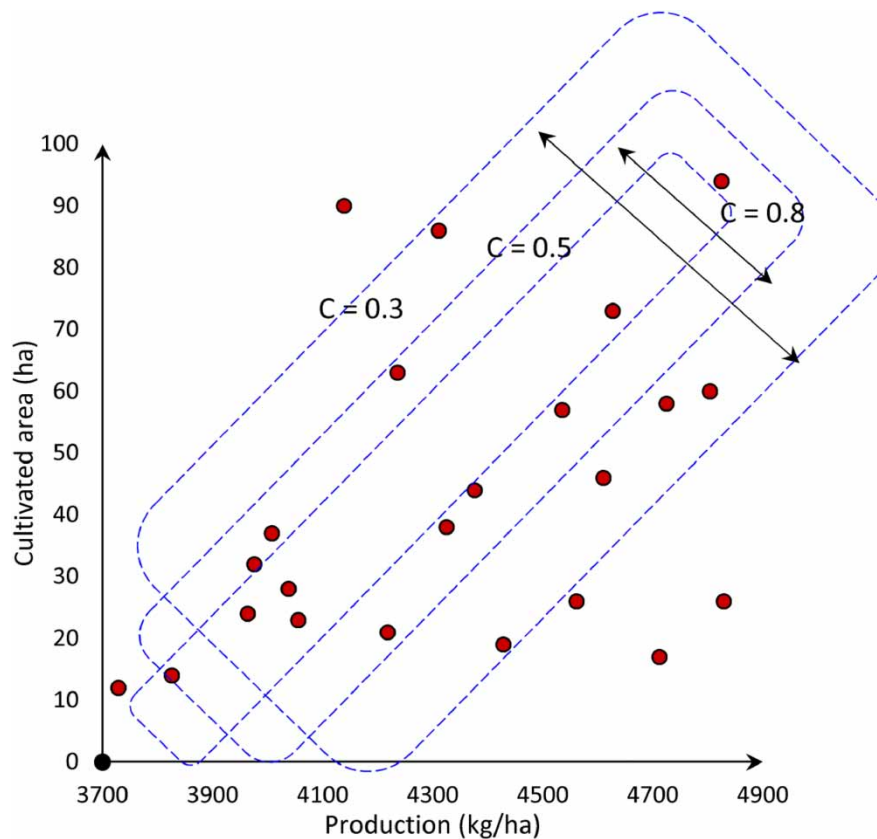


Figure 5 | Correlation between cultivated area and yield production.

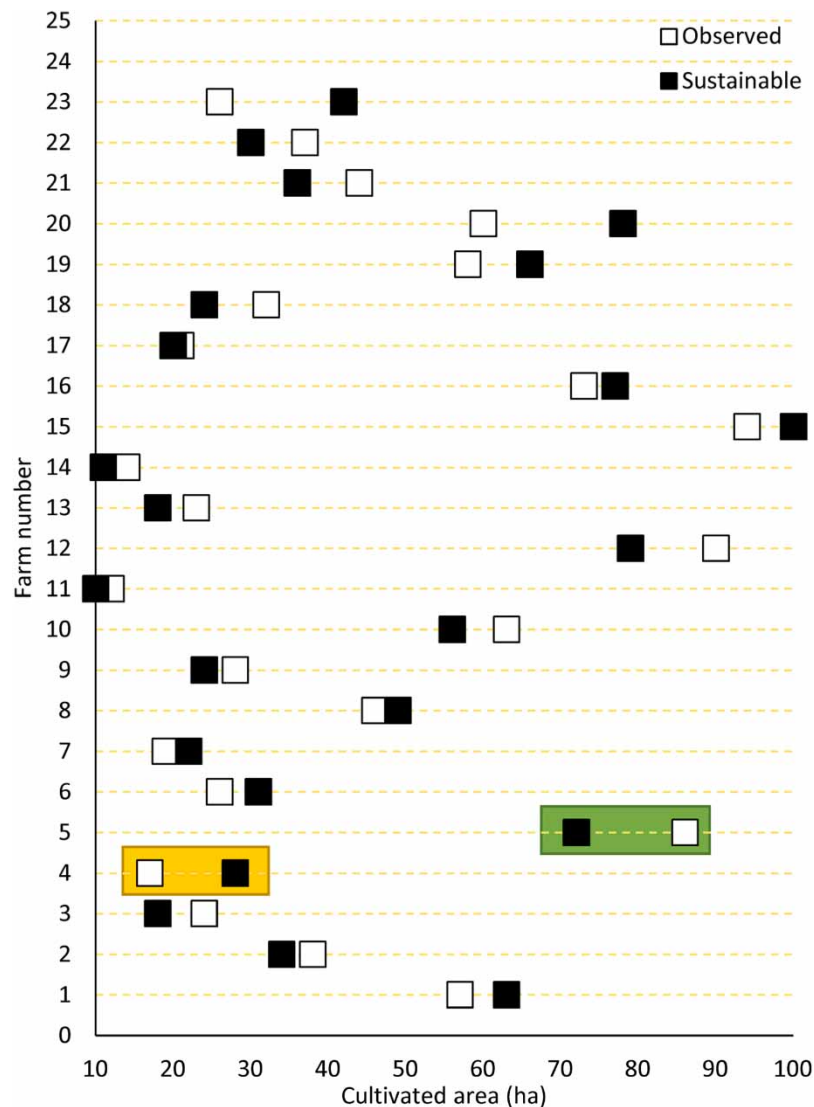


Figure 6 | Observed and sustainable cultivated area.

Managerial factors

Agricultural yield mainly depends on land and precipitation, which are vital components in the growth and production of food crops. Previous studies highlighted that there is a direct relationship between crop growth and land-water patterns. In this study, in addition to water, four parameters of cultivated area, harvest index, length of growth period and agricultural costs were investigated by Pearson's correlation coefficient. Figure 7 shows the role of sustainable model development through machine learning algorithms in improving this correlation (Liu *et al.* 2022). Graphical results showed that the correlation coefficient between harvest index and production was more than other parameters. The term 'harvest index' is used in agriculture to quantify the yield of a crop species versus the total amount of biomass that has been produced. According to the obtained results, the proposed model cannot increase the correlation of the harvest index with production in the sustainability strategy (Table 2). In sustainable conditions, only two farms have a correlation higher than 0.8 compared to the observed conditions. Irrigation water and cultivation costs are the next priorities in relation to production. According to Figure 7 and Table 2, optimal conditions can significantly increase the correlation between production, irrigation and cost. The length of the growth period had the least role in the production and the sustainable model had a limited effect on the improvement.

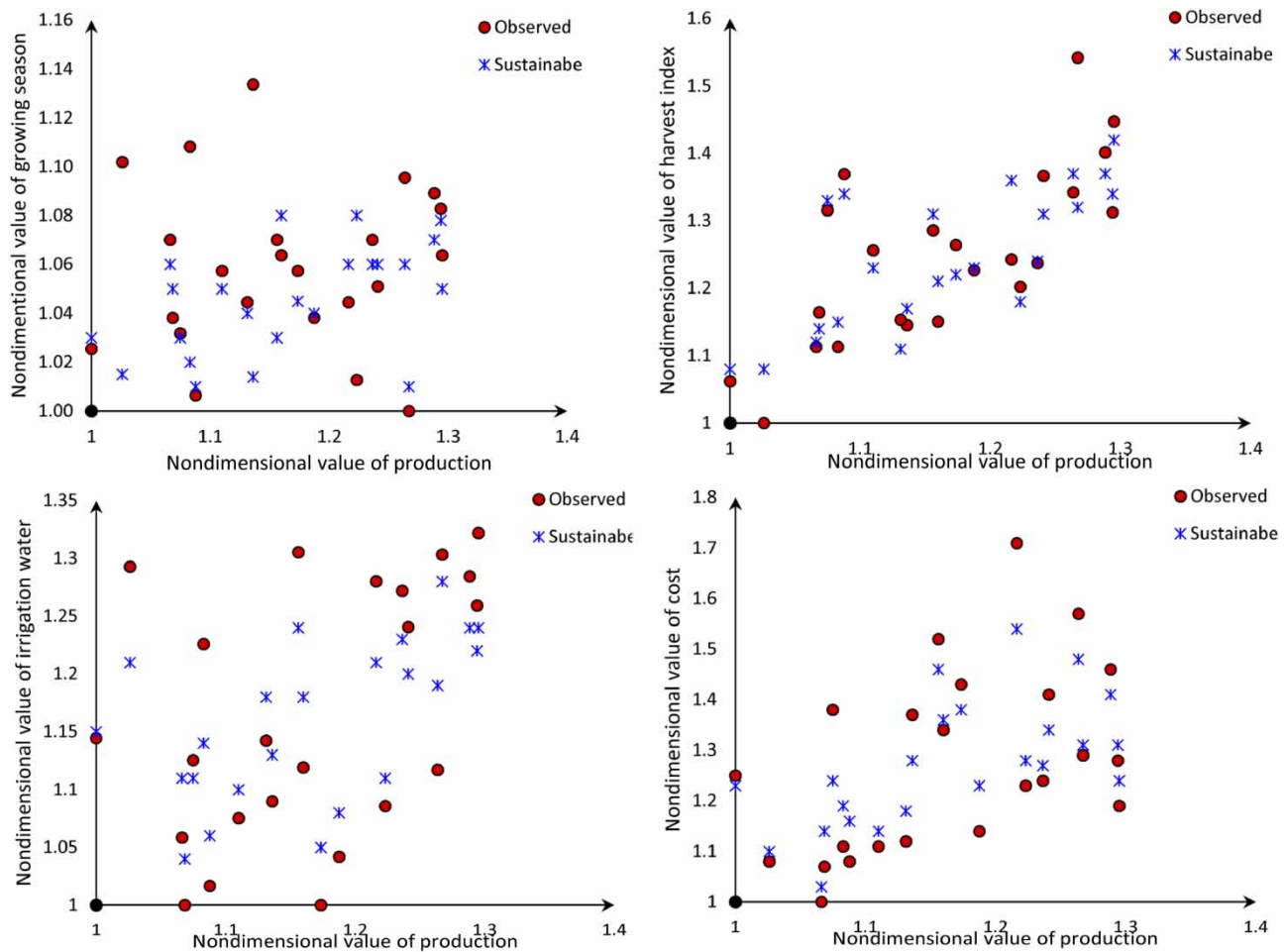


Figure 7 | Correlation between non-dimensional values of managerial factors and production.

Table 2 | Farm classification according to the correlation coefficients between production and managerial factors

Managerial factor	Pcc	Number of farms		Managerial factor	Pcc	Number of farms	
		Observed	Sustainable			Observed	Sustainable
Irrigation water	<0.8	8	12	Harvest index	80	13	15
	0.5–0.8	14	10		50	7	6
	0.3–0.5	1	1		30	3	2
	<0.3	0	0		0	0	0
Cultivation cost	<0.8	12	17	Growing season	80	3	4
	0.5–0.8	7	4		50	6	7
	0.3–0.5	4	2		30	8	8
	<0.3	0	0		0	6	4

SUMMARY AND CONCLUSION

Prediction of yield is one of the most significant and critical processes in accuracy planning in the development of sustainable decision-making models. Smart farming based on sustainable decision-making makes it possible to conduct agriculture more effectively and efficiently for food security. Results showed that machine learning algorithms find solutions which are able to

improve managerial strategies without the involvement of manpower. Correlations between the production and its factors were the criteria used to find the solution of extreme machine learning. It is possible to achieve stable conditions in food security if there is timely and accurate information on decision variables. Irrigation water, planting costs and harvest index were controllable criteria to increase production. In the future research, the role of other crops in the cultivation pattern should be emphasized on the increase of economic indicators. The result of this process in the short term will identify the different dimensions of the water, land and food nexus.

FUNDING

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

REFERENCES

- Abolpour, B., Javan, M. & Karamouz, M. 2017 [Water allocation improvement in river basin using adaptive neural fuzzy reinforcement learning approach](#). *Appl. Soft Comput.* **7**, 265–285. doi:10.1016/j.asoc.2005.02.007.
- Bali, N. & Singla, A. 2022 [Emerging trends in machine learning to predict crop yield and study its influential factors: A survey](#). *Arch. Comput. Methods Eng.* **29** (1), 95–112.
- Bhimavarapu, U., Battineni, G. & Chintalapudi, N. 2023 [Improved optimization algorithm in LSTM to predict crop yield](#). *Computers* **12** (1), 10. <https://doi.org/10.3390/computers12010010>.
- Chen, H., Huang, J. J. & McBean, E. 2020 [Partitioning of daily evapotranspiration using a modified Shuttleworth-Wallace model, random forest and support vector regression, for a cabbage farmland](#). *Agric. Water Manage.* **228**, 105923. <https://doi.org/10.1016/j.agwat.2019.105923>.
- Chithambarathanu, M. & Jeyakumar, M. K. 2023 [Survey on crop pest detection using deep learning and machine learning approaches](#). *Multimedia Tools Appl.* <https://doi.org/10.1007/s11042-023-15221-3>
- Chlingaryan, A., Sukkarieh, S. & Whelan, B. 2018 [Machine learning approaches for crop yield prediction and nitrogen status estimation in precision agriculture: A review](#). *Comput. Electron. Agric.* **151** (November 2017), 61–69. <https://doi.org/10.1016/j.compag.2018.05.012>.
- Elavarasan, D. & Vincent, P. M. D. 2020 [Crop yield prediction using deep reinforcement learning model for sustainable agrarian applications](#). *IEEE Access* **8**, 86886–86901. doi:10.1109/ACCESS.2020.2992480.
- Elavarasan, D. & Vincent, P. M. D. R. 2021 [A reinforced random forest model for enhanced crop yield prediction by integrating agrarian parameters](#). *J. Ambient Intell. Hum. Comput.* **12**, 10009–10022. <https://doi.org/10.1007/s12652-020-02752-y>.
- Elavarasan, D., Vincent, D. R., Sharma, V., Zomaya, A. Y. & Srinivasan, K. 2018 [Forecasting yield by integrating agrarian factors and machine learning models: A survey](#). *Comput. Electron. Agric.* **155**, 257–282. <https://doi.org/10.1016/j.compag.2018.10.024>.
- Frausto-Solis, J., Gonzalez-Sanchez, A., Larre, M., 2009 A new method for optimal cropping pattern. In: *MICAI 2009: Advances in Artificial Intelligence*. MICAI 2009. *Lecture Notes in Computer Science* (Aguirre, A. H., Borja, R. M. & García, C. A. R., eds). Springer, Berlin, Heidelberg. doi:10.1007/978-3-642-05258-3_50.
- Govindan, R. & Al-Ansari, T. 2019 [Simulation-based reinforcement learning for delivery fleet optimisation in CO2 fertilisation networks to enhance food production systems](#). *Comput Aided Chem. Eng.* **46**, 1507–1512. <https://doi.org/10.1016/B978-0-12-818634-3.50252-6>.
- Karmaoui, A., El Jaafari, S. & Chaachouay, H. 2023 [A bibliometric review of geospatial analyses and artificial intelligence literature in agriculture](#). *GeoJournal*. <https://doi.org/10.1007/s10708-023-10859-w>
- Klompenburg, T., Kassahun, A. & Catal, C. 2020 [Crop yield prediction using machine learning: A systematic literature review](#). *Comput. Electron. Agric.* **177**, 105709.
- Lalehzari, R. & Kerachian, R. 2021 [Developing a framework for daily common pool groundwater allocation to demands in agricultural regions](#). *Agric. Water Manage.* **241**, 106278.
- Lalehzari, R., Boroomand-Nasab, S., Moazed, H. & Haghighi, A. 2020 [Simulation–optimization modelling for water resources management using NSGAII-OIP and MODFLOW](#). *Irrig. Drain.* **69** (3), 317–332. <https://doi.org/10.1002/ird.2424>.
- Lamba, V. & Dhaka, V. S. 2014 Wheat yield prediction using artificial neural network and crop prediction techniques (a survey). *Int. J. Res. Appl. Sci. Eng. Technol.* **2**, 330–341.
- Li, S. & Liu, Z. 2022 [Scheduling uniform machines with restricted assignment](#). *Math. Biosci. Eng.* **19** (9), 9697–9708. doi: 10.3934/mbe.2022450.
- Li, M., Fu, Q., Singh, V. & Liu, D. 2018 [An interval multi-objective programming model for irrigation water allocation under uncertainty](#). *Agric. Water Manage.* **196**, 24–36.

- Liu, X., Kong, M., Tong, D. & Zeng, X. 2022 Property rights and adjustment for sustainable development during post-productivist transitions in China. *Land Use Policy* **122**, 106379. doi:10.1016/j.landusepol.2022.106379.
- Milan, S. G., Roozbahani, A. & Banihabib, M. E. 2018 Fuzzy optimization model and fuzzy inference system for conjunctive use of surface and groundwater resources. *J. Hydrol.* **566**, 421–434.
- Mirajkar, A. B. & Patel, P. L. 2016 Multiobjective two-phase fuzzy optimization approaches in management of water resources. *J. Water Resour. Plann. Manage.* [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0000682](https://doi.org/10.1061/(ASCE)WR.1943-5452.0000682).
- Mokhtar, A., Al-Ansari, N. & El-Ssawy, W. 2023 Prediction of irrigation water requirements for green beans-based machine learning algorithm models in arid region. *Water Resour. Manage.* **37**, 1557–1580. <https://doi.org/10.1007/s11269-023-03443-x>.
- Neto, J. R., de Souza, Z. M., de Medeiros Oliveira, S. R., Kölln, O. T., Ferreira, D. A., Carvalho, J. L. N., Braunbeck, O. A. & Franco, H. C. J. 2017 Use of the decision tree technique to estimate sugarcane productivity under edaphoclimatic conditions. *Sugar Tech* **19**, 662–668. <https://doi.org/10.1007/s12355-017-0509-7>.
- Pantazi, X. E., Moshou, D., Alexandridis, T., Whetton, R. L. & Mouazen, A. M. 2016 Wheat yield prediction using machine learning and advanced sensing techniques. *Comput. Electron. Agric.* **121**, 57–65. <https://doi.org/10.1016/j.compag.2015.11.018>.
- Patrício, D. I. & Rieder, R. 2018 Computer vision and artificial intelligence in precision agriculture for grain crops: A systematic review. *Comput. Electron. Agric.* **153**, 69–81. <https://doi.org/10.1016/j.compag.2018.08.001>.
- Pimenov, D. Y., Bustillo, A. & Wojciechowski, S. 2023 Artificial intelligence systems for tool condition monitoring in machining: Analysis and critical review. *J. Intell. Manuf.* **34**, 2079–2121. <https://doi.org/10.1007/s10845-022-01923-2>.
- Rezaei, F., Safavi, H. R. & Zekri, M. 2017 A hybrid fuzzy-based multi-objective PSO algorithm for conjunctive water use and optimal multi-crop pattern planning. *Water Resour. Manage.* **31** (4), 1139–1155.
- Sepahvand, R., Safavi, H. R. & Rezaei, F. 2019 Multi-objective planning for conjunctive use of surface and ground water resources using genetic programming. *Water Resour. Manage.* **33**, 2123–2137. <https://doi.org/10.1007/s11269-019-02229-4>.
- Singh, A. R., 2023 Crop yield prediction and climate change impact assessment using machine learning technology in agriculture. In: *Information and Communication Technology for Competitive Strategies (ICTCS 2022). Lecture Notes in Networks and Systems*, Vol 615 (Kaiser, M. S., Xie, J. & Rathore, V. S., eds). Springer, Singapore. https://doi.org/10.1007/978-981-19-9304-6_33
- Virnodkar, S. S., Pachghare, V. K. & Patil, V. C. 2020 Remote sensing and machine learning for crop water stress determination in various crops: A critical review. *Precis. Agric.* **21**, 1121–1155. doi:10.1007/s11119-020-09711-9.

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